

mielke-geometrodynamics: residuals

equations measured 2026-09-05T13:17:30Z against the model of 2026-09-01

Corrected (3)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_128 EQ0378 (was)	128	■ 0.136	$D\,D\,\vartheta^{\alpha}{}_{\beta}=R_{\beta}{}^{\alpha}{}_{\gamma}{}^{\delta}\vartheta^{\gamma}{}_{\delta}\wedge\vartheta^{\alpha}{}_{\beta}=0,\quad D\,\vartheta^{\alpha}{}_{\beta}=R_{\beta}{}^{\alpha}{}_{\gamma}{}^{\delta}\vartheta^{\gamma}{}_{\delta}\wedge\vartheta^{\alpha}{}_{\beta}=0,$	$DD\vartheta^{\alpha}=R_{\beta}{}^{\alpha}{}_{\gamma}{}^{\delta}\vartheta^{\gamma}{}_{\delta}=0,\quad D\,D\,\Pi_{\alpha}=-R_{\alpha}{}^{\beta}{}_{\gamma}{}^{\delta}\Pi_{\beta}\wedge\vartheta^{\gamma}{}_{\delta}=\pm\frac{i}{2}\ell^2\left(e_{\alpha}{}^{\beta}\Pi^{\beta}_{\gamma}\right)\wedge\Sigma_{\beta}\cong0.$	$DD\vartheta^{\alpha}=R_{\beta}{}^{\alpha}{}_{\gamma}{}^{\delta}\vartheta^{\gamma}{}_{\delta}=0,\quad D\,D\,\Pi_{\alpha}=-R_{\alpha}{}^{\beta}{}_{\gamma}{}^{\delta}\Pi_{\beta}\wedge\vartheta^{\gamma}{}_{\delta}=\pm\frac{i}{2}\ell^2\left(e_{\alpha}{}^{\beta}\Pi^{\beta}_{\gamma}\right)\wedge\Sigma_{\beta}\cong0.$
mielke-geometrodynamics_128 EQ0378 (now)	128	—	preserve the term Pontryagin index for the classification of real associated vector bundles (\textsc{Mayer} \& \textsc{Drechsler} 1977). Nevertheless, the denotation of \textsc{Belavin}	<i>preserve the term Pontryagin index for the classification of real associated vector bundles (Mayer&Drechsler1977). Nevertheless, the denotation of Belavin</i>	$DD\vartheta^{\alpha}=R_{\beta}{}^{\alpha}{}_{\gamma}{}^{\delta}\vartheta^{\gamma}{}_{\delta}=0,\quad D\,D\,\Pi_{\alpha}=-R_{\alpha}{}^{\beta}{}_{\gamma}{}^{\delta}\Pi_{\beta}\wedge\vartheta^{\gamma}{}_{\delta}=\pm\frac{i}{2}\ell^2\left(e_{\alpha}{}^{\beta}\Pi^{\beta}_{\gamma}\right)\wedge\Sigma_{\beta}\cong0.$
basis: inferred / ink					
mielke-geometrodynamics_134 EQ0402 (was)	134	■ 0.042	$\begin{array}{ll} q: & \underline{\vartheta^{\alpha}{}_{\beta}}\,\Psi_{\parallel}(\vartheta)=\underline{\vartheta^{\alpha}{}_{\beta}}\,\Psi_{\parallel}(\vartheta), \\ \pm p: & \underline{\Pi_{\alpha}}\,\Psi_{\parallel}(\vartheta)=-i\ell^2\frac{\delta}{\delta\underline{\vartheta}_{\alpha}}\Psi_{\parallel}(\vartheta), \end{array}$	$q: \quad \underline{\vartheta^{\alpha}{}_{\beta}}\,\Psi_{\parallel}(\vartheta)=\underline{\vartheta^{\alpha}{}_{\beta}}\,\Psi_{\parallel}(\vartheta), \\ \pm p: \quad \underline{\Pi_{\alpha}}\,\Psi_{\parallel}(\vartheta)=-i\ell^2\frac{\delta}{\delta\underline{\vartheta}_{\alpha}}\Psi_{\parallel}(\vartheta),$	$q: \quad \underline{\vartheta^{\alpha}{}_{\beta}}\,\Psi_{\parallel}(\vartheta)=\underline{\vartheta^{\alpha}{}_{\beta}}\,\Psi_{\parallel}(\vartheta), \\ \pm p: \quad \underline{\Pi_{\alpha}}\,\Psi_{\parallel}(\vartheta)=-i\ell^2\frac{\delta}{\delta\underline{\vartheta}_{\alpha}}\Psi_{\parallel}(\vartheta),$
mielke-geometrodynamics_134 EQ0402 (now)	134	—	$\begin{array}{ll} q: & \underline{\vartheta^{\alpha}{}_{\beta}}\,\Psi_{\parallel}(\vartheta)=\underline{\vartheta^{\alpha}{}_{\beta}}\,\Psi_{\parallel}(\vartheta), \\ \pm p: & \underline{\Pi_{\alpha}}\,\Psi_{\parallel}(\vartheta)=-i\ell^2\frac{\delta}{\delta\underline{\vartheta}_{\alpha}}\Psi_{\parallel}(\vartheta), \end{array}$	$q: \quad \underline{\vartheta^{\alpha}{}_{\beta}}\,\Psi_{\parallel}(\vartheta)=\underline{\vartheta^{\alpha}{}_{\beta}}\,\Psi_{\parallel}(\vartheta), \\ \pm p: \quad \underline{\Pi_{\alpha}}\,\Psi_{\parallel}(\vartheta)=-i\ell^2\frac{\delta}{\delta\underline{\vartheta}_{\alpha}}\Psi_{\parallel}(\vartheta),$	$q: \quad \underline{\vartheta^{\alpha}{}_{\beta}}\,\Psi_{\parallel}(\vartheta)=\underline{\vartheta^{\alpha}{}_{\beta}}\,\Psi_{\parallel}(\vartheta), \\ \pm p: \quad \underline{\Pi_{\alpha}}\,\Psi_{\parallel}(\vartheta)=-i\ell^2\frac{\delta}{\delta\underline{\vartheta}_{\alpha}}\Psi_{\parallel}(\vartheta),$
basis: inferred / ink					
mielke-geometrodynamics_314 EQ0972 (was)	314	■ 0.153	$\stackrel{\circ}{D}\alpha:=\stackrel{\circ}{d}\alpha-\omega^g\wedge\alpha-\alpha\wedge\stackrel{\circ}{\omega}$	$\stackrel{(s)}{D}\alpha:=\stackrel{(o)}{d}\alpha-\omega^g\wedge\alpha-\alpha\wedge\stackrel{\circ}{\omega}$	$\stackrel{(o)}{D}\alpha:=\stackrel{(o)}{d}\alpha-\omega^g\wedge\alpha-\alpha\wedge\stackrel{\circ}{\omega}$
mielke-geometrodynamics_314 EQ0972 (now)	314	—	$\overset{(1)}{D}\alpha:=\overset{(0)}{d}\alpha-\omega^g\wedge\alpha-\alpha\wedge\omega^g$	$\overset{(1)}{D}\alpha:=\overset{(0)}{d}\alpha-\omega^g\wedge\alpha-\alpha\wedge\omega^g$	$\overset{(o)}{D}\alpha:=\overset{(o)}{d}\alpha-\omega^g\wedge\alpha-\alpha\wedge\stackrel{\circ}{\omega}$
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
mielke-geometrodynamics_155 E09472 (7.3.11)	155	■ 0.682 ● C +23	\stackrel{+}{\Omega} = 0 \Longleftrightarrow R_{ij} = 0 \Longleftrightarrow R_{ij} = \Lambda' g_{ij}.	(not rendered)	
mielke-geometrodynamics_278 E08857 (12.4.4)	278	■ 0.925 ● C +3	\left.\mathbf{\mathbb{C}}_n \right.\mathscr{G}^A \cong n\lfloor(D\underline{\tau}^A\right),	(not rendered)	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Low confidence

Identifier	Page	Conf.	LaTeX source	Rendered	Image
mielke-geometrodynamics_101 EQ0293 (4.5.4) [conf 0.052]	101	■ 0.052 ● N +0	$\xi \downharpoonleft L \equiv \left(\xi \downharpoonleft \vartheta^\alpha \right) \wedge \frac{\partial L}{\partial \vartheta^\alpha} + \left(\xi \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi}.$	$\xi \downharpoonleft L \equiv \left(\xi \downharpoonleft \vartheta^\alpha \right) \wedge \frac{\partial L}{\partial \vartheta^\alpha} + \left(\xi \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi}.$	$\xi \downharpoonleft L \equiv \left(\xi \downharpoonleft \vartheta^\alpha \right) \wedge \frac{\partial L}{\partial \vartheta^\alpha} + \left(\xi \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi}.$
mielke-geometrodynamics_47 EQ0064 (2.8.6) [conf 0.054]	47	■ 0.054 ● N -2	$\begin{aligned} & \stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \operatorname{Tr} \Omega \quad (=0) \\ & \stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \operatorname{Tr}(\Omega \wedge \Omega) - \operatorname{Tr}(\Omega) \wedge \operatorname{Tr}(\Omega) \} \\ & \quad \left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right) \\ & \stackrel{*}{\pi}(\gamma_k) = 0 \text{ for } k \geq 3. \end{aligned}$	$\begin{aligned} & \stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \operatorname{Tr} \Omega \quad (=0) \\ & \stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \operatorname{Tr}(\Omega \wedge \Omega) - \operatorname{Tr}(\Omega) \wedge \operatorname{Tr}(\Omega) \} \\ & \quad \left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right) \\ & \stackrel{*}{\pi}(\gamma_k) = 0 \text{ for } k \geq 3. \end{aligned}$	$\begin{aligned} & \stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \operatorname{Tr} \Omega \quad (=0) \\ & \stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \operatorname{Tr}(\Omega \wedge \Omega) - \operatorname{Tr}(\Omega) \wedge \operatorname{Tr}(\Omega) \} \\ & \quad \left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right) \\ & \stackrel{*}{\pi}(\gamma_k) = 0 \text{ for } k \geq 3. \end{aligned}$
mielke-geometrodynamics_133 EQ0400 (6.4.19) [conf 0.060]	133	■ 0.060 ● S +1	$\stackrel{(\pm)}{\mathcal{G}}_\alpha := \underline{D}^{(\pm)} \Pi_\alpha = \underline{D}^{(\pm)} \underline{A}_\alpha^{(\pm)} = 0,$	$\stackrel{(\pm)}{\mathcal{G}}_\alpha := \underline{D}^{(\pm)} \Pi_\alpha = \underline{D}^{(\pm)} \underline{A}_\alpha^{(\pm)} = 0,$	$\stackrel{(\pm)}{\mathcal{G}}_\alpha := \underline{D}^{(\pm)} \Pi_\alpha = \underline{D}^{(\pm)} \underline{A}_\alpha^{(\pm)} = 0,$
mielke-geometrodynamics_229 EQ0718 (10.9.4) [conf 0.066]	229	■ 0.066 ● N +0	$\begin{aligned} & \Sigma_\alpha := e_\alpha \downharpoonleft L_\Psi - (e_\alpha \downharpoonleft \Psi) \wedge \frac{\partial L_\Psi}{\partial \Psi} - (e_\alpha \downharpoonleft \bar{\Psi}) \wedge \frac{\partial L_\Psi}{\partial \bar{\Psi}} \\ & \quad - (e_\alpha \downharpoonleft D\Psi) \wedge \frac{\partial L_\Psi}{\partial D\Psi} - (e_\alpha \downharpoonleft D\bar{\Psi}) \wedge \frac{\partial L_\Psi}{\partial D\bar{\Psi}}; \end{aligned}$	$\Sigma_\alpha := e_\alpha \downharpoonleft L_\Psi - (e_\alpha \downharpoonleft \Psi) \wedge \frac{\partial L_\Psi}{\partial \Psi} - (e_\alpha \downharpoonleft \bar{\Psi}) \wedge \frac{\partial L_\Psi}{\partial \bar{\Psi}} - (e_\alpha \downharpoonleft D\Psi) \wedge \frac{\partial L_\Psi}{\partial D\Psi} - (e_\alpha \downharpoonleft D\bar{\Psi}) \wedge \frac{\partial L_\Psi}{\partial D\bar{\Psi}};$	$\begin{aligned} \Sigma_\alpha &:= e_\alpha \downharpoonleft L_\Psi - (e_\alpha \downharpoonleft \Psi) \wedge \frac{\partial L_\Psi}{\partial \Psi} - (e_\alpha \downharpoonleft \bar{\Psi}) \wedge \frac{\partial L_\Psi}{\partial \bar{\Psi}} \\ &\quad - (e_\alpha \downharpoonleft D\Psi) \wedge \frac{\partial L_\Psi}{\partial D\Psi} - (e_\alpha \downharpoonleft D\bar{\Psi}) \wedge \frac{\partial L_\Psi}{\partial D\bar{\Psi}}; \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Image
mielke-geometrodynamics_131 EQ0391 (6.4.6) [conf 0.066]	131	■0.066 ●	$\begin{aligned} \Sigma_{\alpha} &:= \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ &\quad - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}. \end{aligned}$	$\Sigma_{\alpha} := \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - \left(e_{\alpha} \rfloor R_{\beta}^{\gamma}\right) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}.$	$\Sigma_{\alpha} := \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}.$
mielke-geometrodynamics_168 EQ0529 (7.7.49) [conf 0.099]	168	■0.099 ●	$\begin{aligned} L_{\text{Euler}} &:= \frac{(-1)^{\text{sig}+1}}{2} \text{Tr} \left\{ \Omega \wedge \Omega^{(*)} \right\} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ &= dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(*)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ &\equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^*R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^*\text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}^{\alpha} \wedge {}^*\text{Ric}_{\beta}^{\beta} \\ &\equiv -L_{\text{SKY}} \\ &\quad - 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^*\text{Ric}^{\alpha\beta} - \frac{1}{4} \text{Ric}_{\alpha}^{\alpha} \wedge {}^*\text{Ric}_{\beta}^{\beta} \right) \end{aligned}$	$L_{\text{Euler}} := (-1)^{\text{sig}+1} \text{Tr} \left\{ \Omega \wedge \Omega^{(*)} \right\} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ = dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(*)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ \equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^*R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^*\text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}^{\alpha} \wedge {}^*\text{Ric}_{\beta}^{\beta} \\ \equiv -L_{\text{SKY}} \\ - 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^*\text{Ric}^{\alpha\beta} - \frac{1}{4} \text{Ric}_{\alpha}^{\alpha} \wedge {}^*\text{Ric}_{\beta}^{\beta} \right)$	$L_{\text{Euler}} := (-1)^{\text{sig}+1} \text{Tr} \{ \Omega \wedge \Omega^{(*)} \} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ = dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(*)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ \equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^*R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^*\text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}^{\alpha} \wedge {}^*\text{Ric}_{\beta}^{\beta} \\ \equiv -L_{\text{SKY}} \\ - 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^*\text{Ric}^{\alpha\beta} - \frac{1}{4} \text{Ric}_{\alpha}^{\alpha} \wedge {}^*\text{Ric}_{\beta}^{\beta} \right)$
mielke-geometrodynamics_315 EQ0974 (14.2.15) [conf 0.099]	315	■0.099 ●N +0	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\circ\Theta} = 0 \right],$	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\circ\Theta} = 0 \right],$	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\pi\hat{\Theta} = 0} \right],$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					